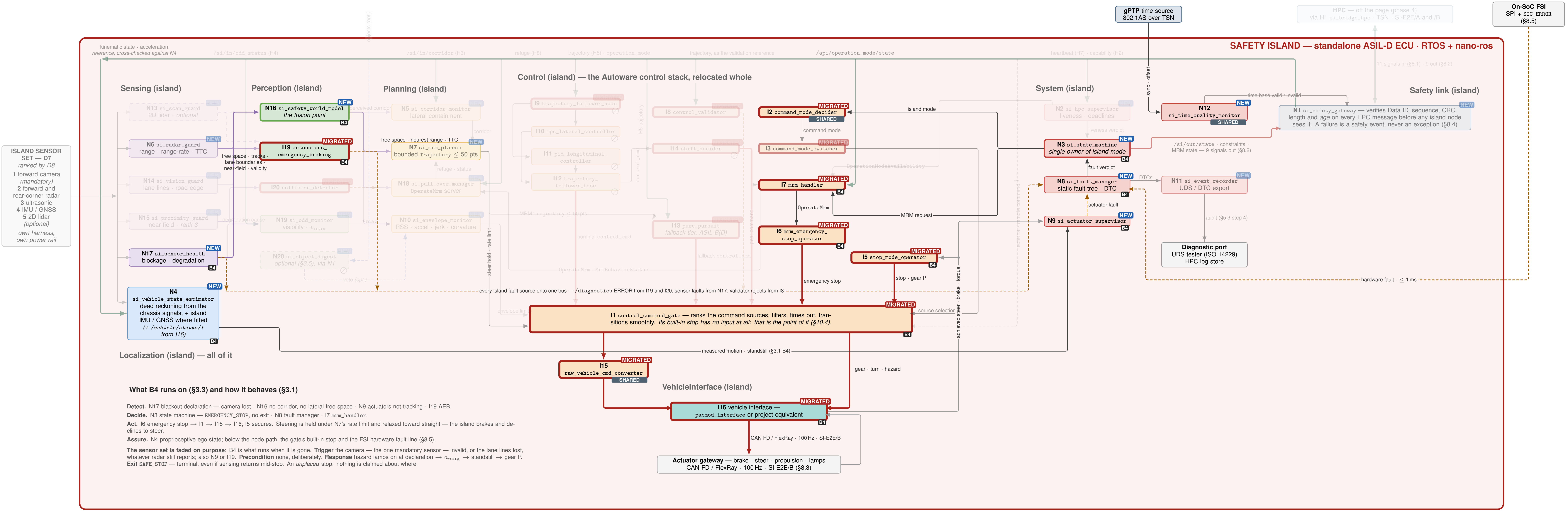
Phase 5 · B4 — Emergency stop when the island is blind

When the island has lost the sensing that B2 and B3 both depend on — the camera — decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre.
The phase-5 drawing, unchanged: what B4 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
objects (opt.)
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
11 signals in (§8.1) · 9 out (§8.2)
sync · offset
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera (mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar (optional)
own harness, own power rail
Sensing (island)
NEW
N13 si_scan_guard
2D lidar · optional
NEW
N6 si_radar_guard
range · range-rate · TTC
NEW
N14 si_vision_guard
lane lines · road edge
NEW
N15 si_proximity_guard
near-field · rank 3
NEW
N17 si_sensor_health
blockage · degradation
B4
Perception (island)
NEW
N16 si_safety_world_model
the fusion point
B4
MIGRATED
I19 autonomous_emergency_braking
B4
free space · tracks · lane boundaries · near-field · validity
MIGRATED
I20 collision_detector
degradation cause
NEW
N19 si_odd_monitor
visibility · $v_{\max}$
NEW
N20 si_object_digest
optional (§3.5), via N1
veto (opt.)
NEW
N4
si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
B4
Localization (island) — all of it
Planning (island)
perceived corridor
NEW
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
NEW
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
NEW
N18 si_pull_over_manager
OperateMrm server
$v_{\max}$
NEW
N10 si_envelope_monitor
RSS · accel · jerk · curvature
steer hold · rate limit
envelope limits
Control (island) — the Autoware control stack, relocated whole
MIGRATED
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
H5 trajectory
control_cmd
MRM Trajectory ≤ 50 pts
nominal control_cmd
MIGRATED
I8 control_validator
MIGRATED
I14 shift_decider
MIGRATED
I13 pure_pursuit
fallback tier, ASIL-B(D)
fallback control_cmd
gear command
OperateMrm · MrmBehaviorStatus
MIGRATED
I2 command_mode_decider
SHARED
command mode
MIGRATED
I3 command_mode_switcher
island mode
OperationModeAvailability
MIGRATED
I7 mrm_handler
B4
OperateMrm
MRM request
MIGRATED
I6 mrm_emergency_stop_operator
B4
emergency stop
MIGRATED
I5 stop_mode_operator
B4
stop · gear P
achieved steer · brake · torque
external / remote command †
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
MIGRATED
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
B4
source selection
measured motion · standstill (§3.1 B4)
MIGRATED
I15
raw_vehicle_cmd_converter
SHARED
gear · turn · hazard
VehicleInterface (island)
MIGRATED
I16 vehicle interface — pacmod_interface or project equivalent
B4
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
System (island)
NEW
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
NEW
N3 si_state_machine
single owner of island mode
B4
fault verdict
NEW
N8 si_fault_manager
static fault tree · DTC
B4
actuator fault
NEW
N9 si_actuator_supervisor
B4
DTCs
NEW
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
/si/out/state · constraints ·
MRM state — 9 signals out (§8.2)
NEW
N12
si_time_quality_monitor
SHARED
time base valid / invalid
Safety link (island)
NEW
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
hardware fault · ≤ 1 ms

What B4 runs on (§3.3) and how it behaves (§3.1)

Detect. N17 blackout declaration — camera lost · N16 no corridor, no lateral free space · N9 actuators not tracking · I19 AEB.

Decide. N3 state machine — `EMERGENCY_STOP`, no exit · N8 fault manager · I7 `mrm_handler`.

Act. I6 emergency stop → I1 → I15 → I16; I5 secures. Steering is held under N7's rate limit and relaxed toward straight — the island brakes and declines to steer.

Assure. N4 proprioceptive ego state; below the node path, the gate's built-in stop and the FSI hardware fault line (§8.5).

The sensor set is faded on purpose: B4 is what runs when it is gone. **Trigger** the camera — the one mandatory sensor — invalid, or the lane lines lost, whatever radar still reports; also N9 or I19. **Precondition** none, deliberately. **Response** hazard lamps on at declaration → a_{emg} → standstill → gear P. **Exit** `SAFE_STOP` — terminal, even if sensing returns mid-stop. An *unplaced* stop: nothing is claimed about where.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED the same Autoware node, migrated

NEW new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the external / remote command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table.

Behaviours each node enables (§3.3)

one fixed slot per behaviour **B4** — **B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

∅ **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

→ HPC-sourced, verified by N1
→ island sensor data
→ control command into the gate
⇢ fault · `/diagnostics`
→ actuator feedback
→ state · mode · decision
⇢ optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B4, blind emergency stop

full colour — named for B4 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B4's path by §3.1, but not named in §3.3

faded — not used by B4. An edge is drawn at the weaker of its two ends. Only B4's own tag is drawn on each box.